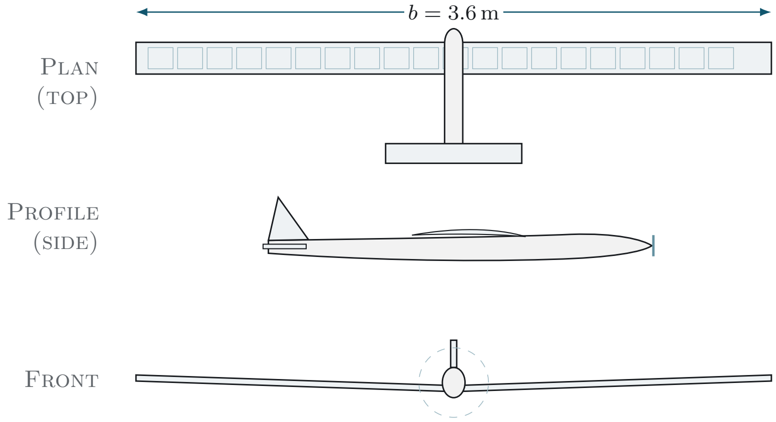
Technical Engineering Report

Aerospace · Power Electronics · Energy Systems

SAM-6 Solar UAV

A Solar-Electric Long-Endurance Unmanned Aerial Vehicle

Systems Engineering, Aerodynamics, Solar Power, Power Electronics, and Energy Analysis



General arrangement, three views (not to scale). The wing upper surface carries the solar array.

Author: Salman K. Alwajaan

Contact: Salman.khalid.alwajaan@gmail.com

Document: Revision 2.0 · 2026

Contents

1 Executive Summary

This report documents the design of SAM-6, a compact unmanned aircraft that draws its flight energy from sunlight. The guiding idea behind the whole project is deliberately simple: instead of fighting the limited energy of batteries with a bigger pack, the airframe is shaped to demand as little power as physically possible. Once the cruise demand is small enough, the modest amount of electricity a wing-mounted solar array can produce becomes more than sufficient to keep the aircraft aloft — and even to refill its battery while it flies.

In numbers, the aircraft spans 3.6 m, carries a slender wing of aspect ratio 14 and area 0.93 m^2, and weighs 3.0 kg at take-off. It cruises at roughly 8 m/s while consuming only about 30 W. Spread across the upper wing surface, an array of about 0.65 m^2 of high-efficiency monocrystalline cells generates close to 123 W at midday under clear skies. The gap between what the sun gives (∼123 W) and what flight costs (∼30 W) is the heart of the design. A 151 Wh lithium-ion pack, wired in a 6S2P arrangement, handles the energy-hungry moments of launch and climb, rides through passing clouds, and holds reserve for the evening.

The endurance claim is framed with care rather than optimism. Adding up the energy harvested over a day and comparing it against the energy spent flying, the two come out roughly equal — so SAM-6 is best described as a strong *daytime* endurance aircraft that hovers near break-even around noon, rather than a machine that flies forever. A target of at least 8 h of solar-assisted flight is set as the figure to demonstrate; flying from sunrise to sunset is within reach on a good day, while round-the-clock operation is treated as a future growth path, not a promise made here.

> **Central idea.** Make the aircraft sip power. A 3.6 m slender wing costs only ∼30 W to cruise, while the array returns ∼123 W at noon — the design wins on efficiency, not on brute power.

2 Problem Statement and Objectives

2.1 Problem Statement

Small battery-powered drones run into the same wall every time: the energy stored in their cells is finite, and most of this class lands after anywhere from a few minutes to a couple of hours. That short clock rules out the missions where staying up matters most — watching a border for a full shift, mapping a wide area in one pass, relaying a radio link, or tracking a slow environmental change. Sun-powered aircraft are an obvious answer, yet many hobby builds chase the idea without doing the engineering underneath it: the lift-to-drag, the load-bearing margins, the day-long energy accounting, and the behavior of the charging electronics are rarely pinned down, so the results are difficult to trust or to improve. SAM-6 is the attempt to close that gap end to end, treating the airframe, the structure, the power path, and the control as one connected problem that another person could rebuild from the numbers given here.

2.2 Objectives and Key Performance Indicators

Objective / KPI	Target	Verification
Daytime endurance	≥ 8 h, solar-augmented	Flight test
Battery-only endurance	≥ 4 h at cruise	Bench + flight
Midday energy balance	Solar power $\geq$ total load	Power logging
Maximum take-off mass	≤ 3.2 kg	Weigh-in
Cruise speed	≈ 8 m/s (6.5 m/s to 14 m/s env.)	Telemetry
Stability margin	Static margin 8 % to 15 % MAC	Analysis + trim

Table 1. Project objectives and how each is verified.

3 Design Assumptions and Reference Conditions

All subsequent analysis rests on the assumptions in Table 2. These are stated explicitly so that results can be reproduced and audited.

Quantity	Value	Basis
Air density ρ	1.225 kg/m^3	ISA sea level
Gravitational acceleration g	9.81 m/s^2	standard
Solar irradiance (clear noon) G	1000 W/m^2	standard test condition
Cell efficiency η_{cell}	23 %	high-efficiency mono
Array system loss factor	0.82	temperature, angle, gaps, wiring
Propeller efficiency η_{prop}	0.70	small low-Re propeller
Drivetrain efficiency η_{drive}	0.80	motor + ESC
Oswald efficiency e	0.85	high-AR planform
Zero-lift drag coeff. C_{D0}	0.030	built-up estimate
Effective sun-hours	5.5 h to 6.5 h	site/seasonal average

Table 2. Reference conditions and modeling assumptions.

4 Aerodynamic Analysis

4.1 The Four Forces and the Cruise Condition

In steady, level flight the four forces balance: lift L equals weight W, and thrust T equals drag D. The weight is

$$W = mg = 3.0\,\mathrm{kg} \times 9.81\,\mathrm{m/s^2} = 29.4\,\mathrm{N}.$$

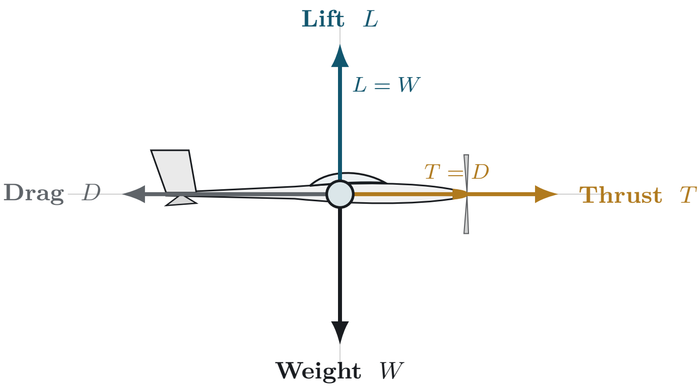


Figure 1. Free-body diagram for steady, level flight. The aircraft is modeled as a point mass at its center of gravity, with the four forces balanced symmetrically: lift equals weight ($L = W$) and thrust equals drag ($T = D$).

4.2 Required Lift Coefficient

From $L = \frac{1}{2}\rho V^2 S\, C_L = W$, the cruise lift coefficient is

$$C_L = \frac{2W}{\rho V^2 S} = \frac{2(29.4)}{1.225\,(8.0)^2\,(0.93)} \approx 0.80.$$

4.3 Drag Polar and Lift-to-Drag Ratio

The drag coefficient combines parasitic and induced contributions:

$$C_D = C_{D0} + \frac{C_L^2}{\pi e\,\mathrm{AR}} = 0.030 + \frac{0.80^2}{\pi(0.85)(14)} \approx 0.030 + 0.017 = 0.047.$$

Hence the cruise lift-to-drag ratio is

$$\frac{L}{D} = \frac{C_L}{C_D} = \frac{0.80}{0.047} \approx 17.$$

Why L/D governs everything. High $L/D \Rightarrow$ low drag $\Rightarrow$ low thrust $\Rightarrow$ low power. The $\sim$30 W cruise is a direct consequence of $L/D \approx 17$, achieved by high aspect ratio (14), an efficient low-Reynolds airfoil (SD7037), and low mass (3.0 kg).

4.4 Reynolds Number

At cruise, with mean chord $c = 0.26\,\mathrm{m}$ and kinematic viscosity $\nu \approx 1.46 \times 10^{-5}\,\mathrm{m}^2/\mathrm{s}$:

$$\mathrm{Re} = \frac{V c}{\nu} = \frac{8.0 \times 0.26}{1.46 \times 10^{-5}} \approx 1.4 \times 10^5,$$

a low-Reynolds regime that motivates the SD7037 airfoil selection.

4.5 Aerodynamic Summary

Parameter	**Value**
Cruise speed V	$8.0\,\mathrm{m/s}$
Stall speed V_{stall}	$6.6\,\mathrm{m/s}$
Cruise lift coefficient C_L	0.80
Cruise drag coefficient C_D	0.047
Cruise drag force D	$1.73\,\mathrm{N}$
Maximum lift-to-drag ratio	$\sim$17
Reynolds number (cruise)	$\sim 1.4 \times 10^5$

Table 3. Summary of cruise aerodynamic parameters.

5 Propulsion Power Budget

The electrical cruise power is derived from the drag in three steps.
(1) Aerodynamic (mechanical) power to overcome drag:

$$P_{\mathrm{aero}} = D\,V = 1.73 \times 8.0 \approx 13.9\,\mathrm{W}.$$

(2) Electrical power, accounting for propeller and drivetrain losses:

$$P_{\mathrm{elec}} = \frac{P_{\mathrm{aero}}}{\eta_{\mathrm{prop}}\,\eta_{\mathrm{drive}}} = \frac{13.9}{0.70 \times 0.80} \approx 25\,\mathrm{W}.$$

(3) Total cruise power including avionics ($\sim$5 W):

$$P_{\mathrm{cruise}} = 25 + 5 = 30\,\mathrm{W}.$$

Climb power is dominated by the rate of potential-energy gain. For a climb rate V_c,

$$P_{\mathrm{climb}} \approx \frac{W V_c}{\eta_{\mathrm{prop}}\eta_{\mathrm{drive}}} + P_{\mathrm{cruise}},$$

which for a modest climb yields a short-duration peak of roughly 205 W.

Flight phase	**Electrical power**
Cruise (level)	$\sim$30 W
Climb (full throttle, transient)	$\sim$205 W

Table 4. Propulsion power by flight phase.

6 Solar Power System

6.1 Peak Power

The peak array power follows from area, irradiance, cell efficiency, and a system loss factor:

$$P_{\text{solar}} = A\,G\,\eta_{\text{cell}}\,k_{\text{loss}} = 0.65 \times 1000 \times 0.23 \times 0.82 \approx 123\,\text{W}.$$

Factor	Value	Justification
Array area A	$0.65\,\text{m}^2$	wing upper surface
Irradiance G	$1000\,\text{W/m}^2$	clear-sky noon (STC)
Cell efficiency η_{cell}	23 %	high-efficiency mono
Loss factor k_{loss}	0.82	temperature, angle, gaps, wiring
Peak output	∼**123** W	—

Table 5. Solar array peak-power breakdown.

Solar vs. demand. ∼123 W available at midday against a ∼30 W cruise need — about four times the requirement. The surplus (∼93 W) charges the battery in flight.

6.2 Daily Generation Profile

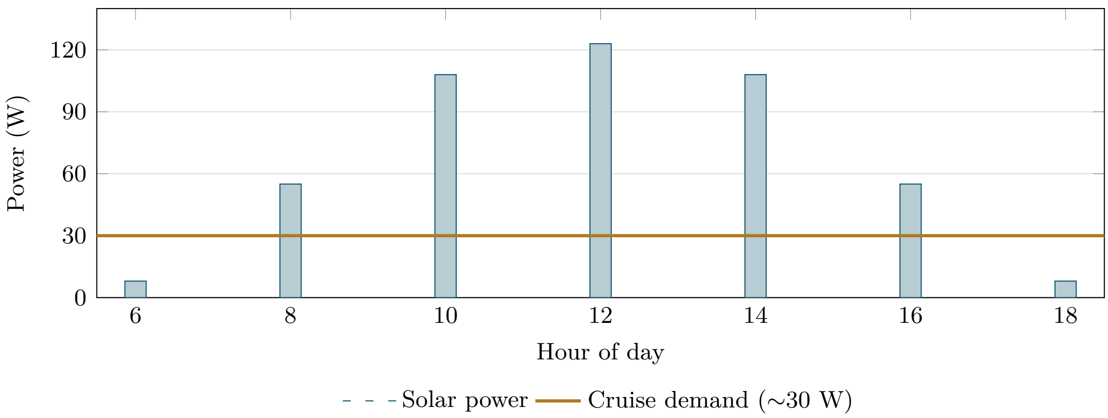


Figure 2. Typical clear-day solar generation versus cruise demand. Generation exceeds demand for most of the day; the area between the curves is energy stored in the battery.

7 Solar Array Wiring: Series versus Parallel

7.1 Single Cell

A single cell produces only ∼0.55 V at its maximum-power point, essentially independent of size; cell area sets the current. Useful voltages are obtained only by combining cells.

7.2 Series and Parallel

In **series**, cell voltages add while current is unchanged (as with batteries stacked in a flashlight). In **parallel**, currents add while voltage is unchanged (as with hoses filling one bucket). Concisely: *series adds volts; parallel adds amps.*

7.3 Chosen Configuration

A buck-type MPPT requires array voltage above the battery's full voltage (25.2 V); the design targets $V_{mp} \approx 33$ V. With ∼0.55 V per cell,

$$N_{\text{series}} = \frac{33}{0.55} \approx 60 \text{ cells.}$$

Array: 60 cells in series	**Value**
Voltage at maximum power V_{mp}	∼33 V
Open-circuit voltage V_{oc}	∼41 V
Current at maximum power I_{mp}	∼3.7 A
Peak power	∼123 W

Table 6. Electrical characteristics of the series string.

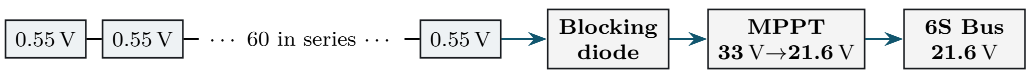


Figure 3. String topology: 60 series cells (∼33 V) feed the MPPT, which steps the voltage down to the 21.6 V bus. Bypass diodes (per cell group) and a string blocking diode provide shading tolerance and reverse protection.

8 Maximum Power Point Tracking (MPPT)

For any level of sunlight and any cell temperature, a solar cell delivers its most power at one particular voltage — push it above or below that point and the output drops. The job of the MPPT is to keep the array parked at exactly that sweet spot, moment by moment, and hand the resulting power to the battery at whatever voltage the bus happens to be. Wiring the panel straight to the battery would do neither: the array would drift off its best point and lose output, and nothing would stop the pack from overcharging. Since the converter neither creates nor destroys power, dropping the voltage from 33 V to 21.6 V must push the current up to compensate:

$$I_{\text{bus}} = \frac{P}{V_{\text{bus}}} = \frac{123}{21.6} \approx 5.7 \text{ A.}$$

At cruise the load draws ∼1.4 A from the bus, so the remaining ∼4.3 A charges the battery near solar noon.

9 Energy Balance and Endurance

Power is instantaneous; energy is its integral over time. Over a representative day:

$$E_{\text{in}} = P_{\text{solar}} \times t_{\text{sun}} = 123 \times (5.5\text{–}6.5) \approx 680\,\text{Wh to } 800\,\text{Wh},$$

$$E_{\text{out}} = P_{\text{cruise}} \times 24\,\text{h} = 30 \times 24 = 720\,\text{Wh}.$$

Quantity	**Value**
Daily solar harvest E_{in}	680 Wh to 800 Wh
Daily consumption E_{out}	∼720 Wh
Net daily balance	approximately neutral
Battery-only endurance	∼4.3 h

Table 7. Daily energy balance.

Reading the result honestly. Around noon the books run in surplus — the aircraft flies and tops up its pack at the same time. Stretched over a full day, the surplus and the deficit very nearly cancel. The fair label is "long daytime endurance" (≥ 8 h), not perpetual flight; reaching the latter would mean a larger wing, more cells, or a lighter structure.

10 Energy Storage: Battery System

Property	**Value**
Configuration	6S2P lithium-ion
Nominal voltage	21.6 V
Full / cut-off voltage	25.2 V / 19.2 V
Stored energy (usable)	∼151 Wh (128 Wh usable)
Battery-only endurance	∼4.3 h

Table 8. Battery pack characteristics.

The label 6S2P unpacks into two stacks of six cells wired in series — six in a row gives $6 \times 3.6\,\text{V} = 21.6\,\text{V}$ — with those two stacks then joined side by side to double the stored charge. It is the very same series-then-parallel trick the solar string uses, only applied to storage instead of generation. Of the full 151 Wh, only about 128 Wh is actually drawn upon, the rest left as a buffer to keep the cells healthy; on that usable slice alone the aircraft can cruise for roughly 4.3 h with the sun switched off entirely.

11 Electrical System Architecture

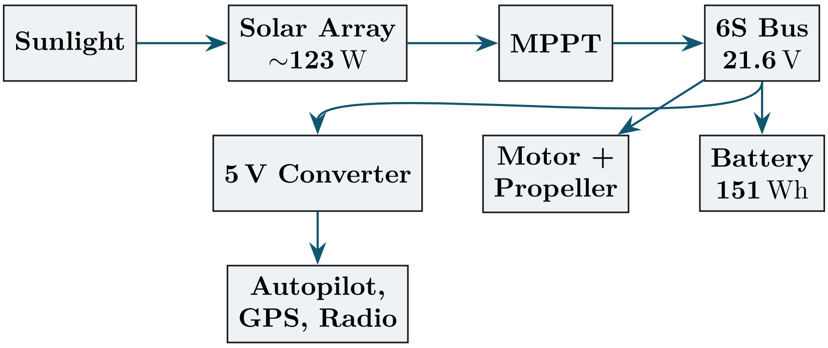


Figure 4. End-to-end power architecture, from sunlight to propulsion and avionics. The battery floats on the 6S bus and absorbs solar surplus.

12 Electronics Hardware: Custom Circuit Boards

The electrical system is carried on three boards laid out by hand in KiCad, each one taken through the program's electrical- and design-rule checks until it passed clean. The renders that follow are shown at a single fixed height, so any screenshot — whatever its proportions — drops into place without disturbing the page.

12.1 Power Distribution Board

The central node at battery voltage. It unites the battery and MPPT output on a single 6S bus and distributes power to the motor/ESC and to a 5 V converter for avionics, with a fuse and decoupling capacitors for protection and clean power.

Power Distribution Board

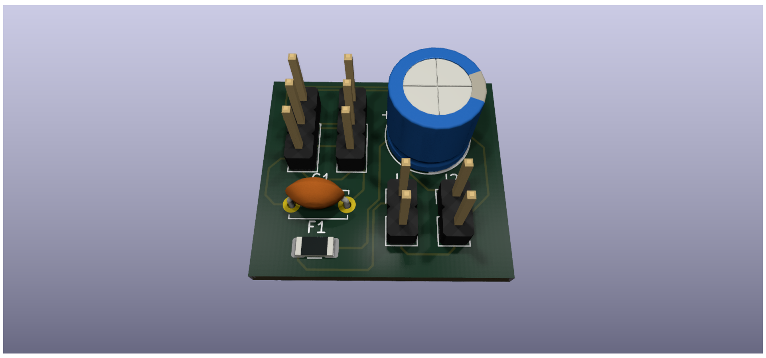


Figure 5. *Power Distribution Board — 3D render (KiCad). Inputs: battery, MPPT output; outputs: ESC, 5 V rail.*

12.2 Solar MPPT Carrier

Conditions and routes solar power. The array input passes a blocking diode and fuse to a high-efficiency MPPT module; its regulated output, with input and output capacitors, feeds the battery bus. A telemetry tap reports voltage and current.

Solar MPPT Carrier

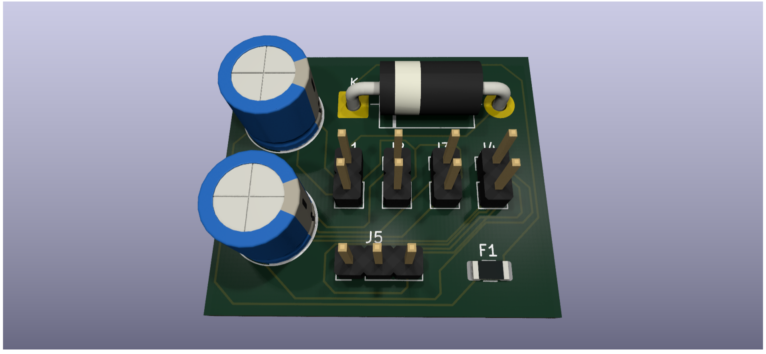


Figure 6. *Solar MPPT carrier — 3D render (KiCad). Solar input → protection → MPPT module → bus output with telemetry sense.*

12.3 Telemetry Interface

The signal hub linking the flight controller, telemetry radio, and I^{2}C sensors. Series resistors protect the UART lines, pull-up resistors support the I^{2}C bus, a status LED indicates power, and a decoupling capacitor cleans the 5 V supply.

Telemetry Interface

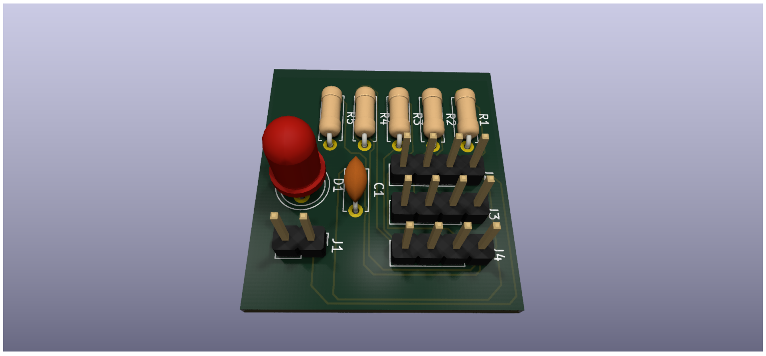


Figure 7. *Telemetry interface — 3D render (KiCad). Links flight controller to radio and I^{2}C sensors with line protection.*

12.4 Airframe CAD

Airframe CAD

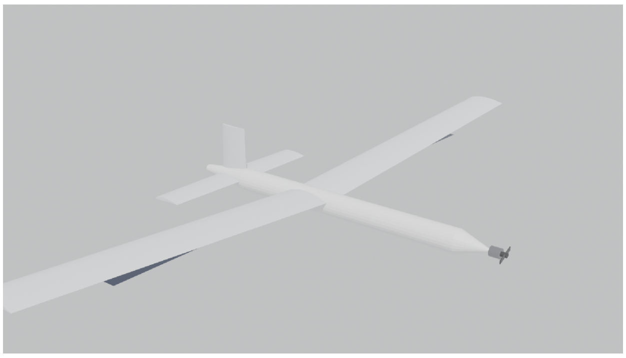


Figure 8. *Airframe CAD render .*

13 Avionics and Power Monitoring

Keeping the aircraft in the air is handled by a Pixhawk-class autopilot running a well-proven flight stack — the kind of code that has flown enough hours to be trusted with the airframe. The bookkeeping of energy is handed off to a separate, smaller brain: an ESP32 that listens to two INA226 sensors over I^2C, one watching the solar feed and one watching the main bus. From their voltage and current readings it works out instantaneous power and tallies the running total of watt-hours in and out, then sends the figures down the radio link. Importantly, the ESP32 is kept clear of anything safety-critical: it neither flies the aircraft nor runs the solar conversion, so a fault in the monitor cannot bring down the flight.

Subsystem	**Voltage**	**Power**	**Function**
Autopilot (Pixhawk)	5 V	∼2.5 W	flight control
GPS + compass	5 V	∼0.5 W	navigation
RC receiver	5 V	∼0.5 W	manual link
Telemetry radio	5 V	∼1.0 W	ground data link
ESP32 + INA226 (×2)	5 V	∼0.5 W	energy monitoring
Avionics total	5 V	∼**5** W	—

Table 9. Avionics power budget on the 5 V rail.

14 Structural Summary

The wing is carried by a single carbon tube, 16 mm across, running along its span. Working out how the lift loads up that tube toward the root gives an ultimate bending moment near 50 N m, which the chosen tube survives with margin to spare. The aircraft is balanced so its center of gravity sits about 30 % of the way back along the mean chord, leaving a static margin of roughly 10 % — enough to be steady in the air without making it sluggish to control. Before anyone flies it, the spar and the joint where the wing meets the fuselage should be checked properly with finite-element analysis rather than hand calculation alone.

15 Consolidated Results

Quantity	Result	Significance
Cruise lift coefficient C_L	0.80	efficient operating point
Cruise lift-to-drag L/D	∼17	low drag
Cruise power	∼30 W	solar-feasible
Climb power (peak)	∼205 W	sizes ESC and wiring
Solar peak power	∼123 W	$\sim 4\times$ cruise need
Daily energy balance	approx. neutral	≥ 8 h endurance
Battery-only endurance	∼4.3 h	reserve
Static margin	∼10 % MAC	stable
Spar margin of safety	positive	structurally safe

Table 10. Consolidated engineering results.

16 Risk Register

ID	Risk	Likeli.	Impact	Mitigation
R1	Cruise power exceeds estimate	Med	High	Conservative C_{D0}; flight-validate; trim for best L/D
R2	Solar harvest below model	Med	High	Oversize array margin; MPPT efficiency $\geq 97\%$
R3	Battery over-discharge	Low	High	19.2 V cut-off; usable energy capped at 128 Wh
R4	Shading loss on array	Med	Med	Bypass diodes per cell group
R5	Spar failure in gust	Low	High	Positive margin; FEA before flight
R6	Stability/CG error	Low	High	CG at 30 % MAC; static margin ∼10%
R7	Custom MPPT immature	Med	Med	COTS module on carrier board for Version 1
R8	Thermal derating of cells	Med	Med	Loss factor 0.82 includes temperature

Table 11. Top engineering risks and mitigations.

17 Limitations and Future Work

- Daytime endurance ($\geq 8\,\mathrm{h}$) is realistic; continuous all-night flight is not promised in Version 1.
- Aerodynamic results are first-principles analytical estimates; computational fluid dynamics and wind-tunnel testing are recommended for validation.
- A custom switching MPPT requires the measured array current–voltage curve and bench verification; Version 1 uses a high-efficiency module on a carrier board.
- Structural margins are estimated by beam theory; finite-element analysis of the spar and wing joint should precede first flight.
- Long-term scale-up toward perpetual flight requires a larger wing, additional cells, or reduced structural mass.

18 References

[1] J. D. Anderson, *Fundamentals of Aerodynamics*, McGraw-Hill.
[2] D. P. Raymer, *Aircraft Design: A Conceptual Approach*, AIAA.
[3] M. Drela, *Flight Vehicle Aerodynamics*, MIT Press.
[4] A. Noth, *Design of Solar Powered Airplanes for Continuous Flight*, ETH Zürich.
[5] M. H. Rashid, *Power Electronics: Circuits, Devices, and Applications*, Pearson.
[6] UIUC Applied Aerodynamics Group, low-Reynolds airfoil data (SD7037).
[7] KiCad EDA documentation, `kicad.org`.

SAM-6 attains at least eight hours of daylight endurance because a 3.6 m high-aspect-ratio wing requires only about 30 W to cruise, while approximately 123 W of solar power simultaneously replenishes the battery.

Prepared by Salman K. Alwajaan · Salman.khalid.alwajaan@gmail.com · Revision 2.0, 2026